

Module 1: Network + Network Fundamentals and building networks

IP Addressing and Subnetting

Task 1:

Class	IP Range	Use case
A	1.0.0.0 – 126.255.255.255	Core routers of a massive ISP or a company like Google's internal backbone network
B	128.0.0.0 – 191.255.255.255	A large university or corporate campus network, like an IIT/ college-wide network
C	192.0.0.0 – 223.255.255.255	A home Wi-Fi router (most default to 192.168.x.x, a private Class C range)
D	224.0.0.0 – 239.255.255.255	IPL live streaming to multiple set-top boxes simultaneously over a cable network using multicast
E	240.0.0.0 – 255.255.255.255	Not used in consumer devices — reserved by IETF for future/ experimental protocols

Task 2:

Network/broadcast/host range for 192.168.10.45/255.255.255.0

Given: IP = 192.168.10.45, Subnet mask = 255.255.255.0 (i.e., /24)

Step 1 — Convert to binary (last octet only, since first 3 octets are fully masked):

- IP: 192.168.10.**00101101** (45 in binary)
- Mask: 255.255.255.**00000000**

Step 2 — Since the mask is /24, the first 3 octets define the network, and the last octet is fully available for hosts.

- **Network address** = 192.168.10.0
- **Broadcast address** = 192.168.10.255
- **Valid host range** = 192.168.10.1 to 192.168.10.254

Task 3:

Subnetting plan for a Zomato-style startup (10.0.0.0/24, 4 subnets)

Step 1 — We need 4 subnets from a /24. Since $2^2 = 4$, we borrow 2 bits from the host portion: new prefix = /24 + 2 = **/26**.

Step 2 — /26 = 255.255.255.**192**, giving blocks of 64 addresses each.

Group	Network Address	Subnet Mask	Usable Host Range
Delivery Partner	10.0.0.0/26	255.255.255.192	10.0.0.1 – 10.0.0.62
Restaurants	10.0.0.64/26	255.255.255.192	10.0.0.65 – 10.0.0.126
Customers	10.0.0.128/26	255.255.255.192	10.0.0.129 – 10.0.0.190
Admins	10.0.0.192/26	255.255.255.192	10.0.0.193 – 10.0.0.254

Task 4:

College Wi-Fi: 172.16.0.0/16, at least 20 subnets

Step 1 — Find how many bits are needed so $2^n \geq 20$:

- $2^5 = 32$ (enough) → **borrow 5 bits**

Step 2 — New prefix length = 16 + 5 = **/21**

Step 3 — Convert /21 to a subnet mask:

- /21 means 21 bits are network bits → third octet uses 5 of its 8 bits for subnetting: 11111000 = 248
- **Subnet mask = 255.255.248.0**

Step 4 — Calculate hosts per subnet:

- Remaining host bits = $32 - 21 = 11$ bits
- Hosts per subnet = $2^{11} - 2 = 2048 - 2 = \mathbf{2046}$ **usable hosts per subnet**

Final answer: Subnet mask **255.255.248.0 (/21)**, giving 32 subnets with **2046 usable hosts each** — comfortably covering 20+ buildings.

Task 5:

Why 192.168.1.300 won't connect

The error is that 300 is not a valid value for an IP address octet — each of the four numbers in an IPv4 address must be between 0 and 255 (since each octet is only 8 bits, max value $2^8 - 1 = 255$). Your friend's device likely has a typo or misconfigured static IP.

A valid address in that same subnet (192.168.1.0/24) would be anything from 192.168.1.1 to 192.168.1.254 — for example, 192.168.1.30 or 192.168.1.100 would both work fine.